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Paleomagnetic results from the southeastern margin of the Ebro Basin (northeastern Spain): evidence for a Tertiary clockwise rotation

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Triassic rocks underlying the Paleogene deposits of the Ebro Basin are exposed along its SE margin.

About 200 samples of Middle Triassic and Paleogene rocks from this margin have been demagnetized and analysed. The Middle Triassic red beds contain one or two secondary components of magnetization. The high unblocking temperature component has a mean direction of $D = 1°$ and $I = 6°$.

The Eocene red beds display a very stable magnetization. Nevertheless some samples have a high-temperature component with the present-day field direction. The Eocene mean direction is $D = 22°$ and $I = 51°$.

The paleomagnetic results show a systematic 20° clockwise rotation relative to the data from the Pyrenean and Cantabrian Ranges and the western and SW part of Iberia.

To explain these results two hypotheses are presented. One implies the rotation of the NE part of Iberia, whereas the other refers to a rigid block rotation mechanism along the Catalan Coastal Range.

1. Introduction

Early paleomagnetic studies on the Carboniferous, Permian and Triassic red beds in the Iberian plate indicated a 35° anticlockwise rotation of this plate relative to Europe (Van der Lingen, 1960; Girdler, 1965; Van der Voo, 1967, 1969; Van Dongen, 1967). Further results obtained by Stauffer and Tarling (1971), Vandenberg (1980) and Schott (1985), confirmed this interpretation. At present, the time-span during which the Iberian Plate rotated, can paleomagnetically be defined as Late Cretaceous although some smaller movements may have occurred earlier. These results agree with offshore geophysical studies, which establish that the formation of oceanic crust in the Bay of Biscay, associated with the southeastwards displacement of Iberia, occurred between Early–Middle Albian and Santonian (Olivet et al., 1980). Despite many geophysical and geological studies (Boillot and Capdevila, 1977; Grimaud et al., 1982; Olivet et al., 1984) no paleomagnetic evidence has been found to support a post-Cretaceous rotation of the Iberian Plate.

The present paper contributes to this task by reporting new paleomagnetic results from the Middle Triassic and Paleogene deposits from the SE margin of the Ebro Basin.

2. Geological setting

The Ebro Basin is located in the NE part of the Iberian Peninsula (Fig. 1a). It formed during the

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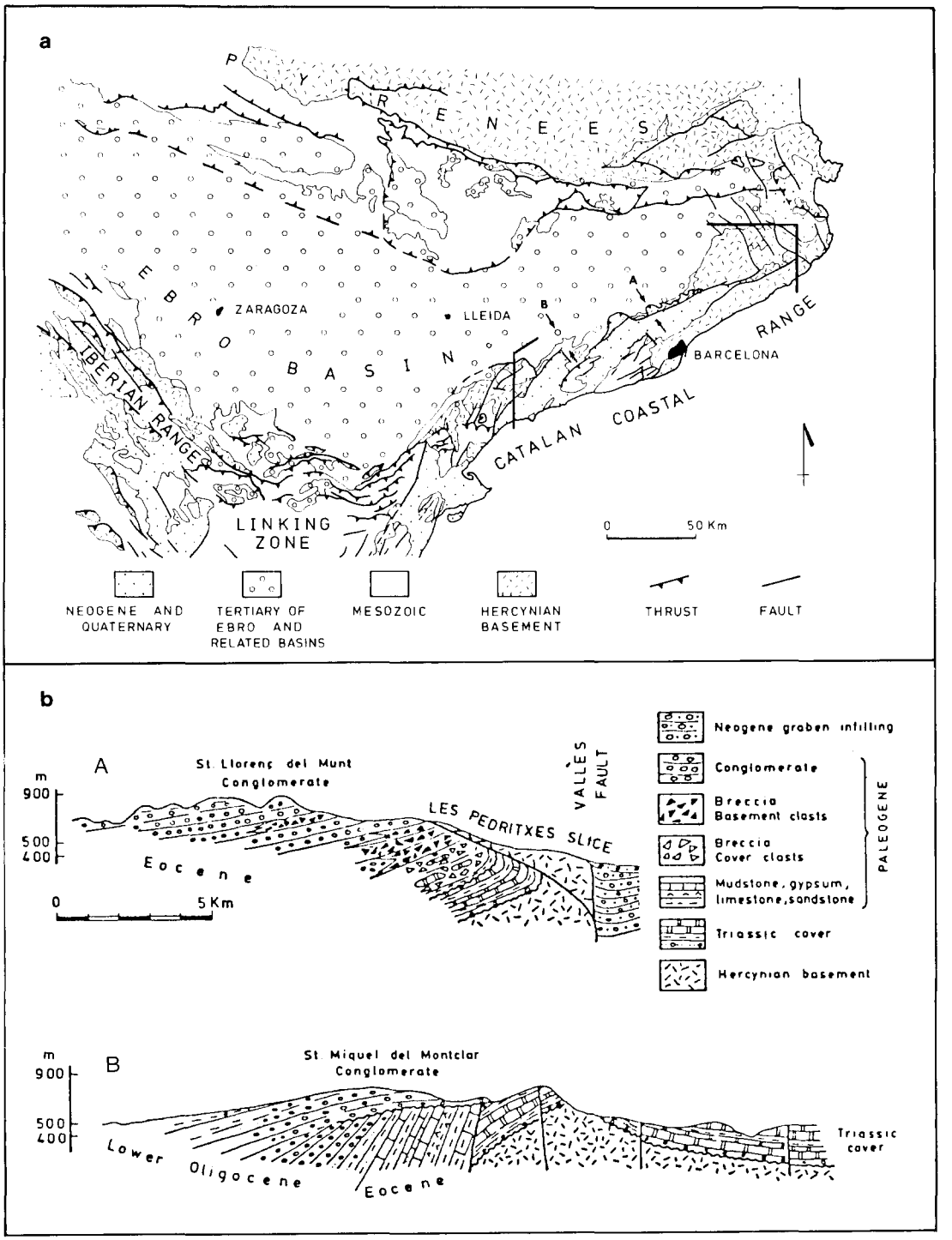
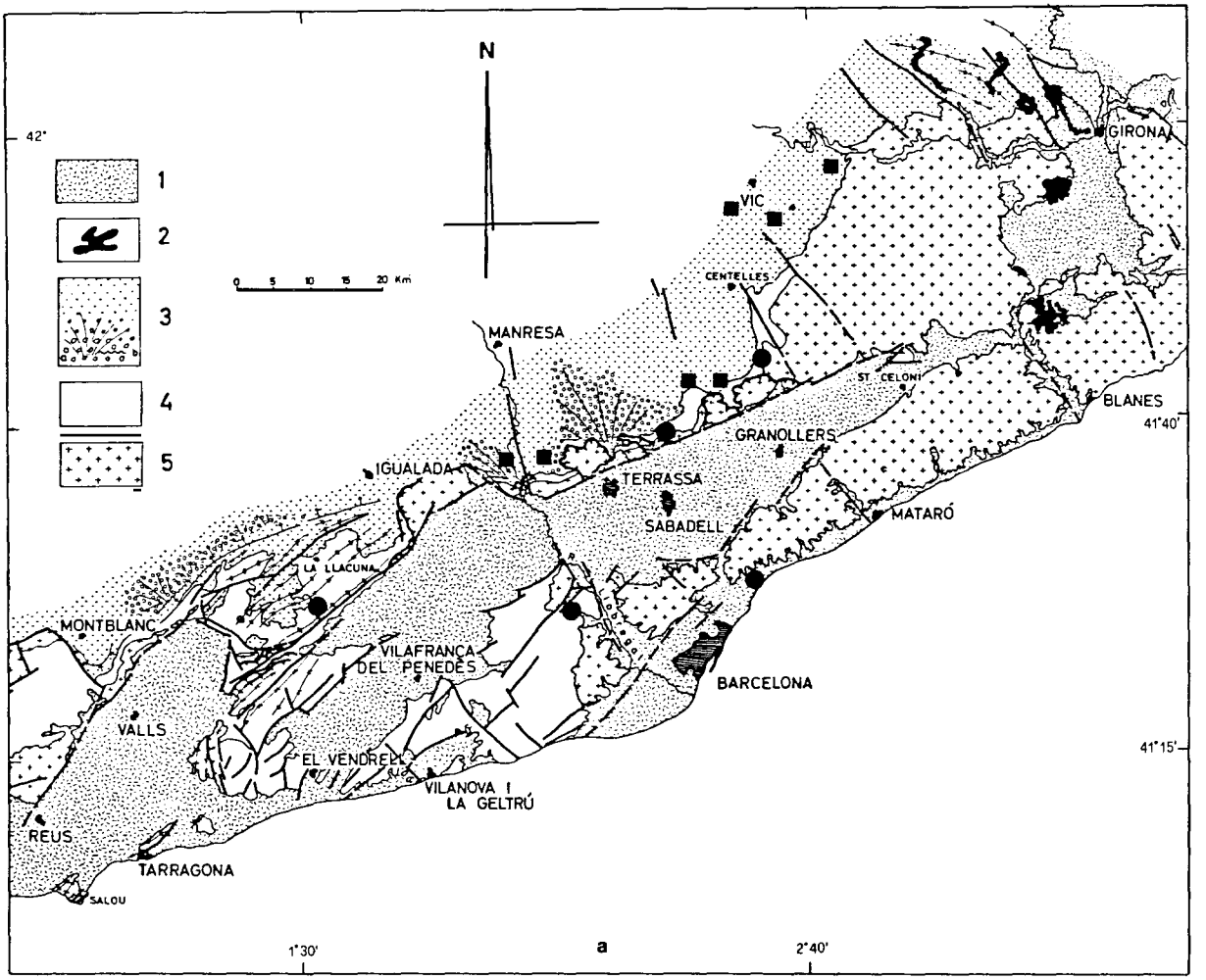


Fig. 1. (a) Generalized map of the study area. The sampled area is enlarged in Fig. 2. (b) Cross-sections through the southeastern margin of the Ebro Basin. They correspond to the arrows marked in the generalized map. Modified from Anadón et al. (1985).



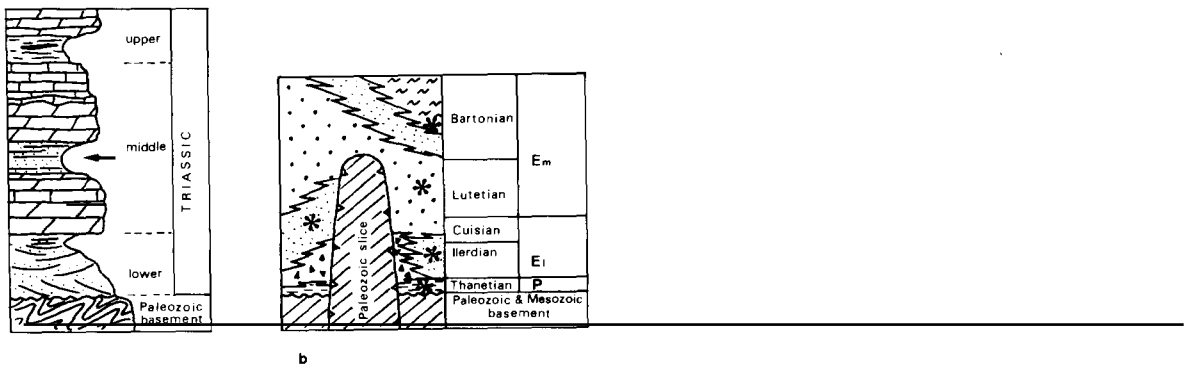


Fig. 2. (a) Geological map of the sampled area. See Fig. 1a for location. 1, Neogene; 2, Quaternary and Neogene volcanic rocks; 3, Paleogene from the Ebro Basin; 4, Mesozoic cover; 5, Paleozoic basement. Modified from Anadón et al. (1979). ●, ■ indicate Triassic and Paleogene sites respectively. (b) Schematic stratigraphic units of the Triassic and Paleogene deposits with the location of the sampled sites. The arrow and the stars indicate sampled formations.

Alpine orogeny as the foreland basin of the Pyrenees, the Iberian Range and the Catalan Coastal Range. In the area studied, SE of the Ebro Basin, coarse- to fine-grained alluvial facies developed from early Eocene to early Oligocene. A widespread transgression affected this area during the Bartonian and was responsible for deposition of carbonates and fine-grained terrigenous rocks (Anadón et al., 1979) (Fig. 2b).

Along the Catalan Coastal Range a Paleozoic basement and its unconformably overlying Mesozoic cover can be distinguished (Figs. 1b, 2a). The basement consists of granitoids and metasedimentary rocks. In the central part of the Range, the Mesozoic cover is mainly composed of Triassic rocks arranged in three lithostratigraphic units (Fig. 2b). The Lower Triassic is represented by red sandstones, mudstones and conglomerates unconformably overlying Paleozoic rocks. The Middle Triassic displays three different levels. In the lower and the upper part there is a carbonatic level. Between them there is a fine-grained terrigenous level characterized by red sandstones and mudstones with some evaporitic rocks. The Upper Triassic rocks are rare in the northern part of the Range because they have been eroded prior to the Paleogene sedimentation. Where present, mudstones, dolomites and evaporitic rocks can be identified.

The principal structures of the Catalan Coastal Range are nearly vertical basement-involved right-stepping faults with an ENE–WSW to NE–SW direction (Guimerà, 1984; Anadón et al., 1985). The most important and intensive Alpine deformation in the Range took place along these major faults. Convergent sinistral slip on these faults resulted in the emplacement of basement slices overthrusting the Paleogene deposits of the Ebro Basin. The Triassic rocks, which are overlain by the Paleogene, were dragged strongly because of the emplacement of the Paleozoic basement slices (Fig. 1b).

The tectonic activity controlled to a great extent the sedimentation during the Paleogene in the Catalan margin of the Ebro Basin. As a consequence, thick alluvial-fan systems were fed from uplifted blocks along the fault-controlled basin boundary and progressive and angular syntectonic unconformities developed (Anadón et al., 1985) (Fig. 1b). The basement-involved faults behaved as normal faults during the Neogene (Julivert et al., 1972–74).

3. Methods

About 90 samples were collected from five sites in fine-grained red sandstones of Middle Triassic age (Fig. 2b). The Tertiary sediments from the Ebro Basin overlying Triassic rocks were also studied. About 100 samples were collected from seven sites in fine- to coarse-grained red sandstones ranging from Late Paleocene to Middle Eocene. Only one site was taken in Middle Eocene marls (Fig. 2b).

Samples were taken in the field with a portable gasoline drill and were cut to 2.5 cm cylinders in the laboratory. A stepwise thermal demagnetization technique was applied using a Schonstedt TSD-1 model oven. The remanent magnetizations of specimens were measured with an ScT three-axis cryogenic magnetometer interfaced with a computer.

The magnetic direction remaining after each step was plotted using Zijderveld vector diagrams (Zijderveld, 1967) and stereographic projections.

In order to monitor any mineralogical changes during heating, bulk susceptibility was measured on a KLY-1 susceptibility bridge during thermal treatment.

4. Analysis of NRM

4.1. Triassic red sandstones

The natural remanent magnetization of the samples have intensities ranging from 4×10^{-4} to 10^{-3} A m^{-1}. The directional changes and the decay of intensity during demagnetization enable us to differentiate two main types of magnetic characteristic.

4.1.1. Type 1

Figure 3 shows the results of thermal demagnetization of a typical specimen having type 1 mag-

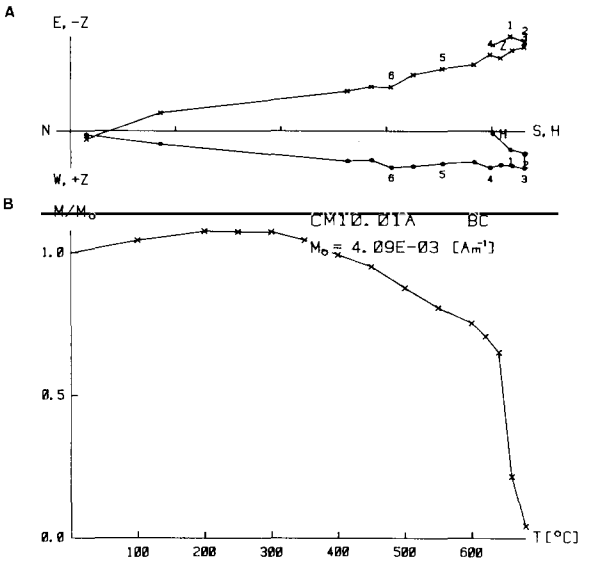


Fig. 3. Thermal demagnetization of Triassic samples having type 1 magnetic characteristics. (A) Zijderveld orthogonal plot with successive vector points during demagnetization. ×, vertical plane; ●, horizontal plane. Numbers refer to every 100 °C. (B) Normalized intensity decay curve.

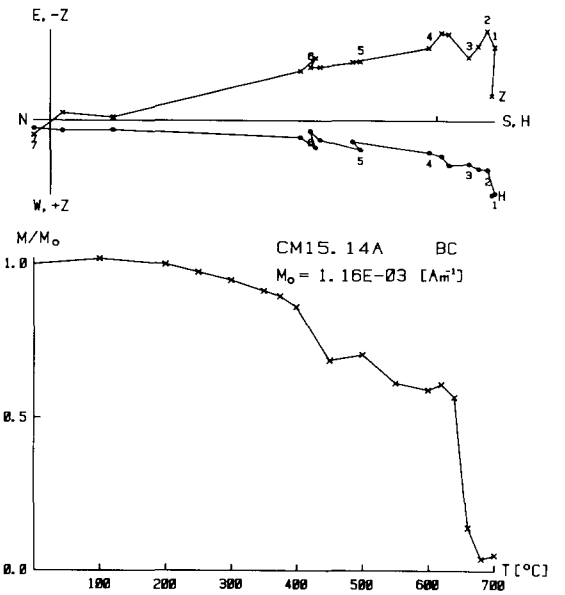


Fig. 4. Thermal demagnetization of green sandstone. Note the two sharp drops, one at 450 °C and another at 680 °C.

netic characteristics. The Zijderveld diagram shows a single component of magnetization with a low unblocking temperature secondary component superimposed which is removed at 250 °C. The primary component has a shallow SSW direction.

The thermal demagnetization curve drops sharply in intensity at 660–680 °C. The intensity decreases very slightly until this temperature is reached (Fig. 3B). In some samples the NRM is composite, consisting of a positive component and an antiparallel negative weaker component.

In some cases there is an initial increase of intensity until 350 °C, followed by a decrease until 660–680 °C, where the intensity drops abruptly. These samples show no change in bulk susceptibility until 550 °C, revealing that the increase in intensity is not due to mineralogical changes and therefore suggesting the presence of antiparallel components prior to heating.

Thermal demagnetization of the green sandstones also shows evidence of a minor low unblocking temperature secondary component superimposed on the main magnetization (Fig. 4). After removal of this secondary component at 200 °C, a component with SSW direction and negative inclination is defined.

The intensity decay shows two thermal demagnetization components, one removed at 450 °C and the other at 680 °C. Titaniferous magnetite and haematite are present in these green sandstones. The directions of characteristic remanent magnetization of the green sandstones are similar to those of the red sandstones described before.

4.1.2. Type 2

Typical samples having type 2 magnetic characteristics are represented in Figs. 5 and 6. The main differences when compared with type 1 are the presence of two secondary components of magnetization and the partial overlap between their unblocking temperature spectra.

The Zijderveld diagram of sample CM16.07B (Fig. 5A) shows three magnetic components with partial overlap but wide windows of non-overlap spectrum. The least stable component is completely removed at 300 °C, whereas the stable component is defined between 500 and 680 °C. The intensity decays within a relatively narrow and high unblocking temperature range and drops sharply at 660–680 °C (Fig. 5B). The resultant

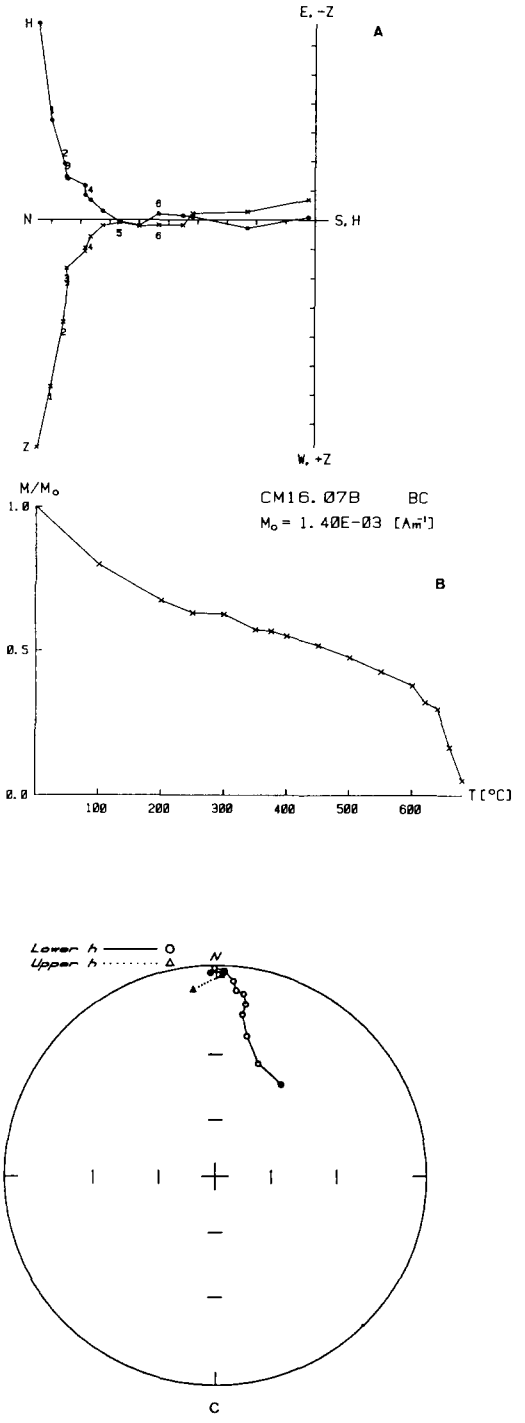


Fig. 5. Thermal demagnetization of a Triassic red sandstone having type 2 magnetic characteristics. (A) and (B) as in Fig. 3. (C) Stereographic projection showing directional changes. The resultant vectors lie along more than one great circle as a consequence of the multicomponent behaviour.

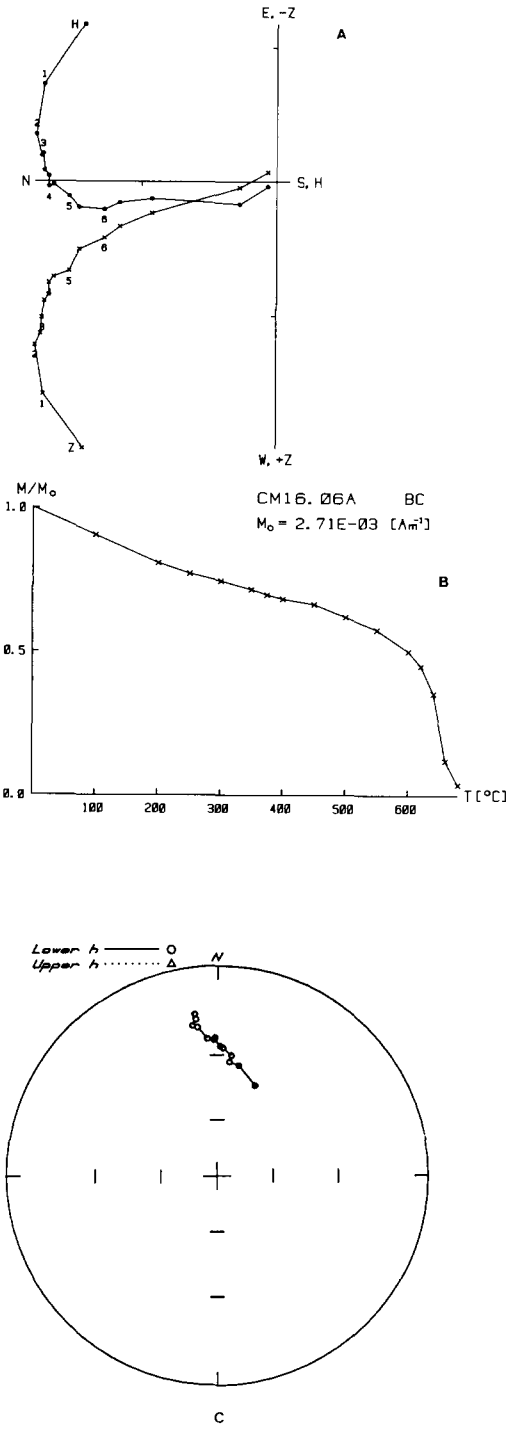


Fig. 6. Thermal demagnetization of sample having type 2 magnetic characteristics. Note the overlap spectrum between the low-temperature components. Other details as in Fig. 3.

vectors lie along more than one great circle (Fig. 5C) owing to the multicomponent behaviour of these samples.

Other samples offer a total overlap spectrum (Fig. 6A), preventing a definition of the low-temperature component of magnetization. In both cases, methods based on converging circles have to be used in order to define these components (Halls, 1978). Vectors associated with the highest-temperature steps lying on the best-fitting straight line were analysed by principal component analysis (Kirschvink, 1980). A detailed study of the remagnetization in these red sandstones is in progress.

4.2. Paleogene formations

Two main groups of magnetic behaviour can be distinguished.

4.2.1. Type 1

Figure 7 shows the results of thermal demagnetization of a fine red sandstone having the first type of magnetic behaviour. The Zijderveld diagram shows evidence for a low unblocking temperature secondary component superimposed on the main one. After removal of this component at 200 °C, the NRM decays throughout a relatively wide unblocking temperature spectrum. From 200 °C onwards a SSW component with relatively steep negative inclination is removed, but the line determined by the resultant vectors does not point to the origin, suggesting that at least another component is still present. In some samples the trajectories of demagnetization bypass the origin in the Zijderveld diagram. Acquisition of viscous magnetization made further measurements impossible. The analysis of this remaining magnetization will be discussed in the next section.

4.2.2. Type 2

The thermal demagnetization of a red sandstone specimen of the second group of magnetic behaviour is shown in Fig. 8. The main differences are the presence of only two magnetic components. A low unblocking temperature component is removed below 100 or 200 °C. At higher temperatures a single component of magnetization directed towards the SSW with rather steep negative inclination is observed. The loss of intensity in these specimens is linear from 200 to 600 °C.

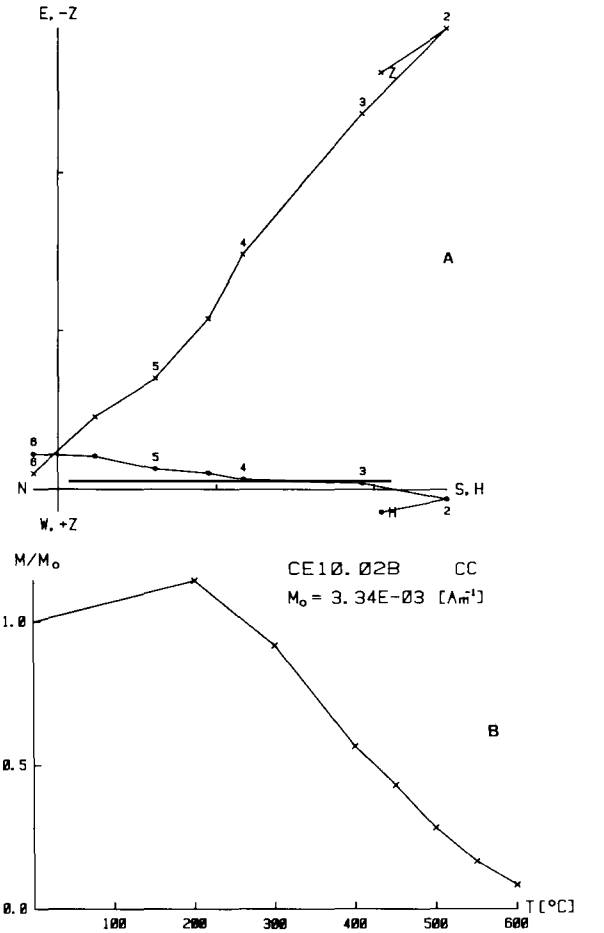


Fig. 7. Thermal demagnetization of an Eocene red sandstone sample showing three magnetic components. The trajectories of demagnetization do not end at the origin. Other details as in Fig. 3.

A few specimens offer minor directional movement above the low unblocking temperature component (Fig. 9). This movement has been reported previously in red beds and may be the result of closely overlapping sub-components (Turner et al., 1985).

The bulk mineralogical changes of red sandstones during heating have been studied by means of normalized initial susceptibility curves. The bulk susceptibility of the red sandstones increases largely upon heating above 400–450 °C. In the argillaceous rocks this increase in susceptibility is

found at 600 °C. In both cases, only when corresponding increase or scattering in magnetization intensity above this temperature were observed were the samples rejected.

The increase in susceptibility in red beds during heating has been reported by many authors, and it is usually related to magnetite production by two different processes: the reduction of haematite (personal communication by Creer in Dunlop (1972)) and the breakdown of some non-ferromagnetic minerals (Dunlop, 1972) such as chamosite (Barrie et al., 1980) or vermiculites (Perarnau and Tarling, 1985).

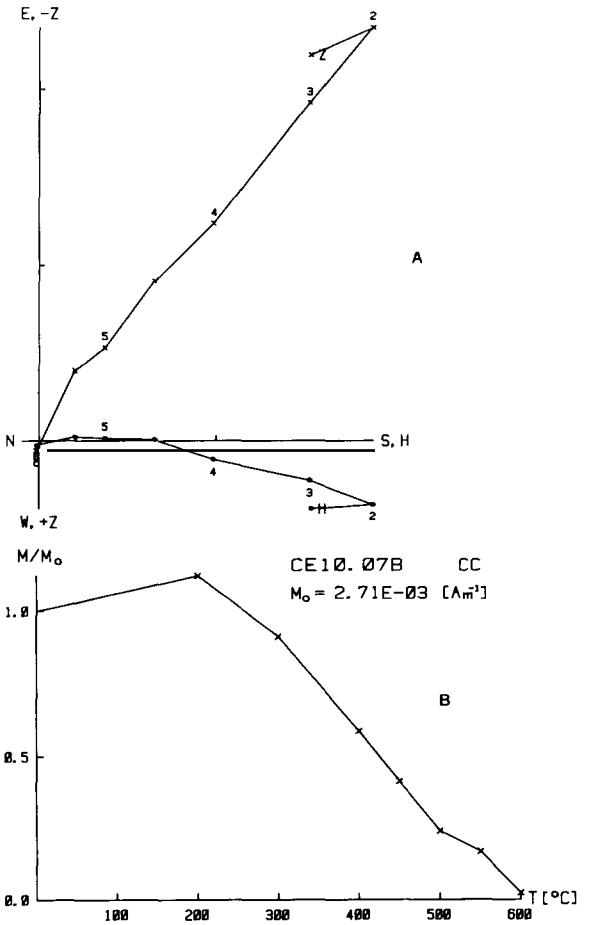


Fig. 8. Thermal demagnetization of an Eocene red sandstone showing two magnetic components. Other details as in Fig. 3.

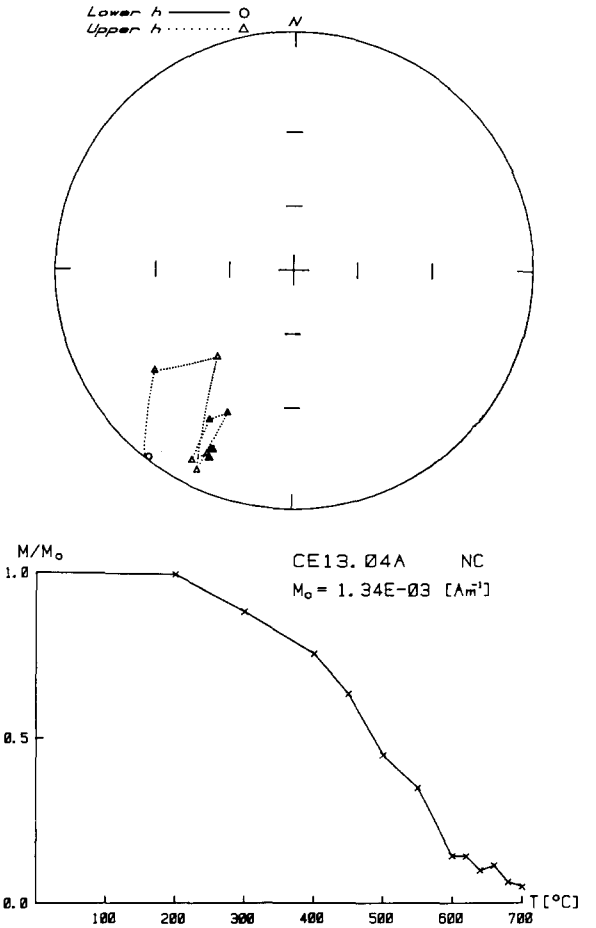


Fig. 9. Thermal demagnetization of an Eocene red sandstone with directional movement during heating as revealed by the stereographic projection. Other details as in Fig. 3.

5. Isothermal remanent magnetization analysis

IRM acquisition curves have been measured on some samples by magnetizing them in progressively increasing fields up to 1 T (Fig. 10). The slope of these curves and the thermal demagnetization curve of the IRM show hematite as the main carrier of magnetization and a small low-coercivity component appears to be present in the red sandstones.

The initial shape of the IRM acquisition curves is slightly convex in the Eocene red sandstones. There is no evidence of full saturation at higher fields because of the high coercivity of the

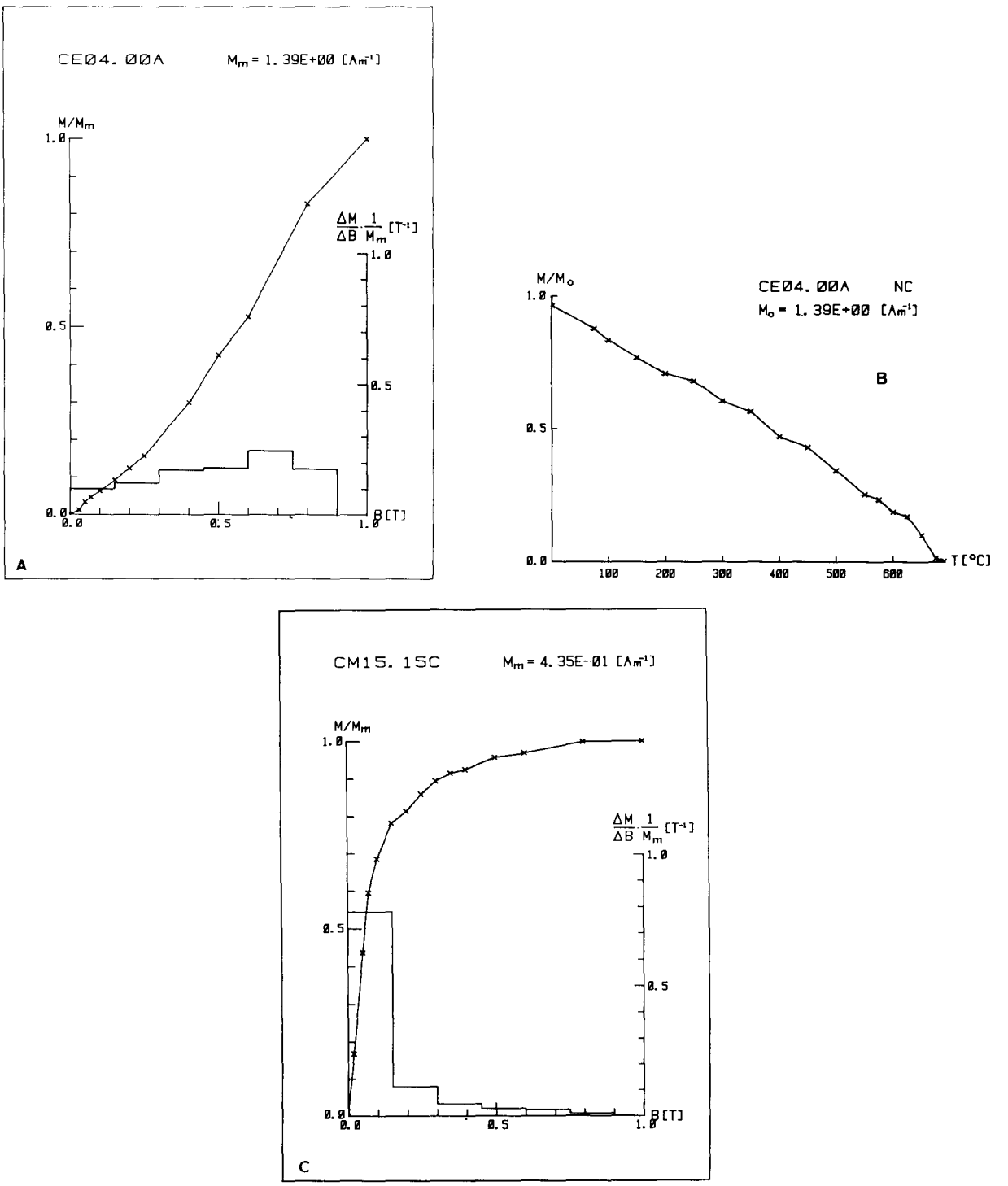


Fig. 10. Isothermal remanent magnetization (IRM). IRM acquisition of Eocene red sandstone with (B) subsequent thermal demagnetization of IRM, (C) IRM acquisition of Triassic green sandstone. Note the flattening off of the slope at 0.8 T, revealing that the low-coercivity component is the main carrier of the IRM.

magnetic component. Thermal demagnetization of IRM shows a broad unblocking temperature spectrum until 680 °C.

This result contrasts with that of the Middle Triassic green sandstones. A steep initial increase and a flattening off of the slope at 0.8 T characterize these rocks, revealing that a low-coercivity component (magnetite) is the main carrier of the IRM.

6. Mean directions

The statistical analysis of the directions (Fisher, 1953) is given for each site in Table I (Triassic sites) and Table II (Eocene sites). The characteristic remanent magnetization directions (ChRM) were determined by principal component analysis (Kirschvink, 1980).

With respect to the Eocene sites, it should be noted that clustering within site is better before tilt correction for some of the sites. At site CE13 (Fig. 11A), the precision parameter k decreases from 39.6 to 19.8 after bedding correction. α_{95} increases from 4.9 to 7°. Similarly, the clustering between sites is better before bedding correction of the Eocene sites (Fig. 11B). Changes in scatter can be attributed to different bedding within site. Assuming that the ChRM found in the Eocene red beds is of primary origin, the improvement of clustering between sites before bedding correction implies that the time of acquisition of the magnetization was after tilting. Syntectonic unconformities which developed at the margin of the basin (Anadón et al., 1986) prove the tilting to be mostly synsedimentary. This is consistent with the geological evolution of the area because the Catalan margin of the Ebro Basin was an active margin

TABLE I

Middle Triassic mean directions

Site	N	Before tilt correction				After tilt correction			
		D	I	k	α_{95}	D	I	k	α_{95}
CM10	20	8.0	27.2	31.4	5.9	0.6	12.5	36.4	5.5
CM15	24	3.4	19.3	26.7	5.8	1.5	7.6	23.7	6.2
CM16	24	359.3	12.0	23.9	6.2	358.8	8.0	23.5	6.2
CM17	10	7.1	3.3	30.4	8.9	7.0	0.7	28.9	9.1
CM18	19	357.4	2.7	46.1	5.0	357.8	4.5	46.7	5.0
Site Mean	5	2.9	12.9	50.3	10.9	1.1	6.3	205.4	5.6

N, Number of samples, D and I, declination and inclination respectively, k precision parameter; α_{95}, radius of the cone of 95% confidence around the mean (Fisher, 1953).

TABLE II

Eocene mean directions (see Table I for explanations)

Site	N	Before tilt correction				After tilt correction			
		D	I	k	α_{95}	D	I	k	α_{95}
CE01	16	29.2	55.1	7.7	14.2	29.3	48.1	7.8	14.2
CE04	17	32.6	46.3	21.4	7.9	28.6	29.4	27.9	6.9
CE05	13	9.6	45.5	11.2	13.0	23.2	45.5	11.2	13.0
CE06	18	20.7	63.1	17.5	8.6	7.7	58.7	15.4	9.1
CE09	15	19.3	53.3	140.1	3.2	6.9	51.8	139.9	3.2
CE10	17	11.0	48.4	739.5	1.3	6.7	47.3	742.1	1.3
CE13	23	21.1	49.8	39.6	4.9	18.8	10.9	19.8	7.0
Site Mean	7	22.5	51.8	114.2	5.7	18.3	42.1	21.1	13.6

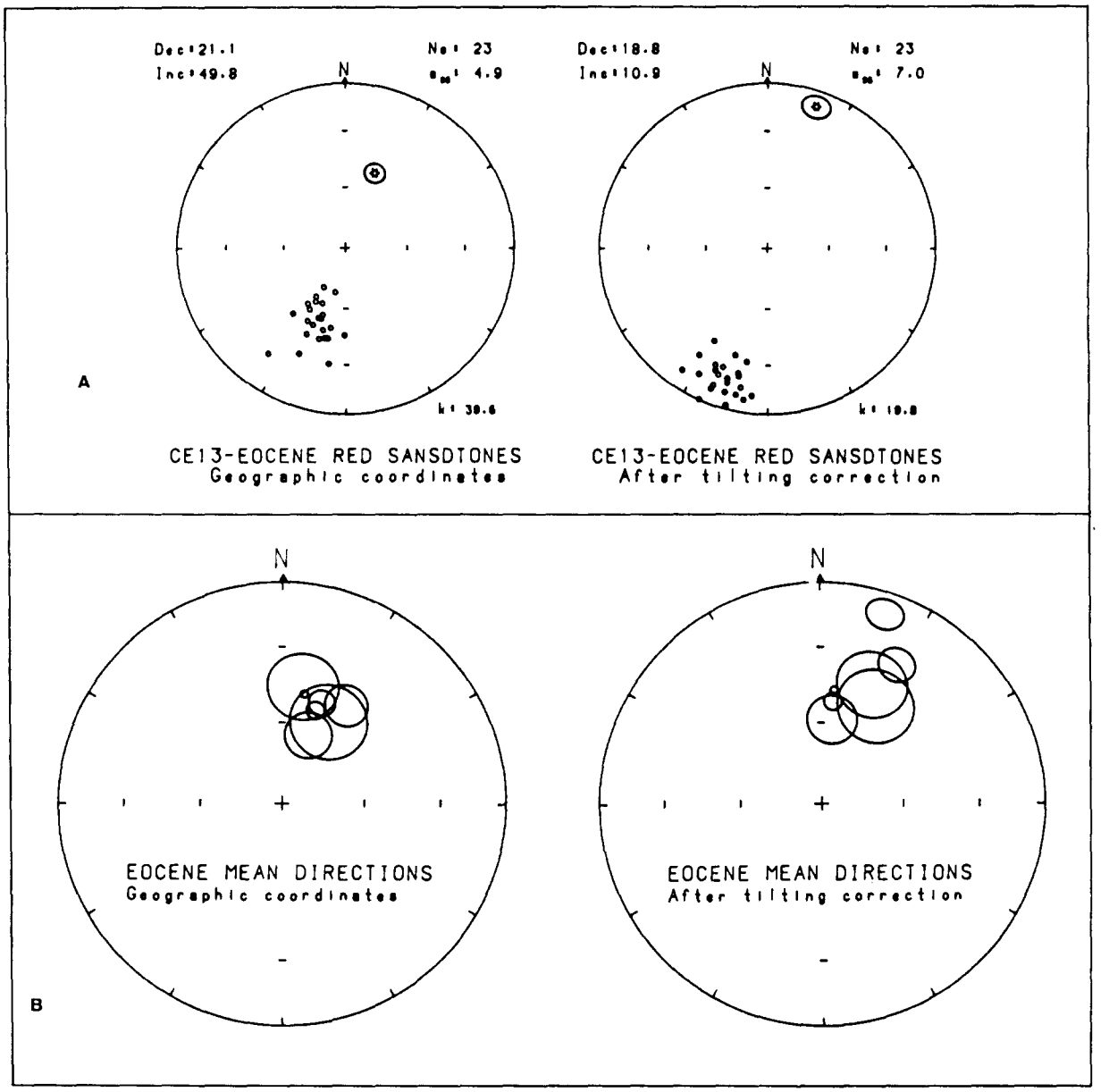


Fig. 11. (A) Stereographic projection of the ChRM vectors for site CE13 (Eocene red sandstones). Note the clustering within site before tilt correction. (B) α_{95} circles for the studied Eocene sites. The clustering between sites is larger before tilt correction, as indicated by the overlapping circles.

during a certain period of time (Anadón et al., 1985). The magnetization in these Eocene red sandstones was acquired after tilting during the Paleogene.

As was described before, the Zijderveld diagrams for the Eocene samples show many cases where the demagnetization straight lines do not lead to the origin (Fig. 7). When stable end-points

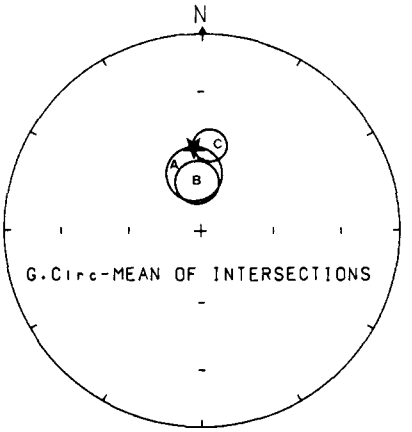


Fig. 12. α_{95} circles of the means of intersection of the great circles of sites CE06 (A), CE09 (B), CE10 (C) before tilt correction, with the present-day field direction (★).

have not been reached, the ChRM directions represent the most stable phase but not the highest-temperature component. In cases where stable end-points have not been reached, it was possible to estimate the direction of the highest-temperature component by fitting the demagnetization trajectories to great circles on a stereographic net and finding the intersection points of these remagnetization circles (Khramov, 1958; Halls, 1978). The demagnetization data were fitted to great circles by least squares (Halls, 1976). The convergence points of great circle intersections and the estimate of the confidence ellipse were found using the method of Bailey and Halls (1984).

The stereographic projection of the mean directions of convergence points with corresponding circles of 95% confidence falls close to the present-day geomagnetic field direction. It suggests a recent age for the highest-temperature component found in some of the Eocene samples (Fig. 12).

7. Discussion and conclusions

Putting together the results obtained in this study with reliable data from the Permian and Triassic of the Iberian Peninsula some conclusions can be drawn.

A clear difference in declination of ~ 20° is observed between the published Permo-Triassic results of Iberia and the Triassic from the studied area (Fig. 13a). If the same exercise is done with Late Cretaceous–Lower Tertiary data, we observe again a 20° difference in declination (Fig. 13b). In fact, recent radiometric results give an age of Upper Cretaceous for the igneous rocks of Portugal used for this comparison (Storetvedt et al., 1987).

The results from Fraga (SE Ebro Basin) (Friend and Cabrera, personal communication, 1987) of late Oligocene, Pertusa (North Ebro Basin) (Turner et al., 1984) of early Miocene and Calata-

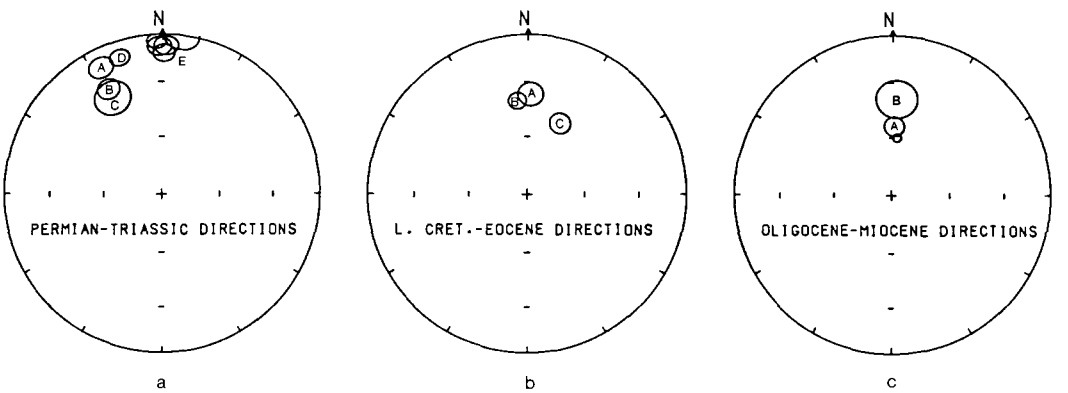


Fig. 13. (a) Permian and Triassic paleomagnetic data from the Cantabrian Range (A: Schott, 1985; B,C: Vandenberg, 1980), Pyrenean Range (D: Schott, 1985) and Catalan Coastal Range (E) (this study). (b) Late Cretaceous–Eocene paleomagnetic data from Monchique (A: Van der Voo, 1969), Lisbon (B: Van der Voo and Zijderveld, 1971) and Eocene from the studied area (C). Note the 20° difference in declination. (c) Miocene and Oligocene paleomagnetic directions from the Ebro Basin (A: Dijksman, 1977; B: Turner et al., 1984; ○: Friend and Cabrera, personal communication, 1987).

yud–Teruel (internal basin in the Iberian Range) (Dijksman, 1977) of middle Miocene age, reveal that no rotation has occurred since then (Fig. 13c).

To explain the systematic 20° clockwise rotation from pre-Oligocene paleomagnetic data, two different models can be put forward. However, both models face severe problems due to geological considerations as well as to the scarce amount and uneven distribution of data.

7.1. Model 1

This model (Fig. 14) is based on the clockwise rotation of the northeastern part of the Iberian Plate relative to the so called 'Stable Meseta'. As a consequence of the convergence between the African and European plates, a continental block including the Ebro Basin, the Catalan Coastal Range and part of the Iberian Range, would be expelled towards the SE due to the movement of the Iberian Plate towards the NNW (Fig. 14).

An analogous model is that proposed by Dewey and Sengör (1979) for the Arabia–Europe intracontinental convergence zone. According to their model, the continental collision, involving irregular margins, led to the lateral escape of the Anatolian plate, as a buoyant continental sliver, from the suture zone into the western Mediterranean tract.

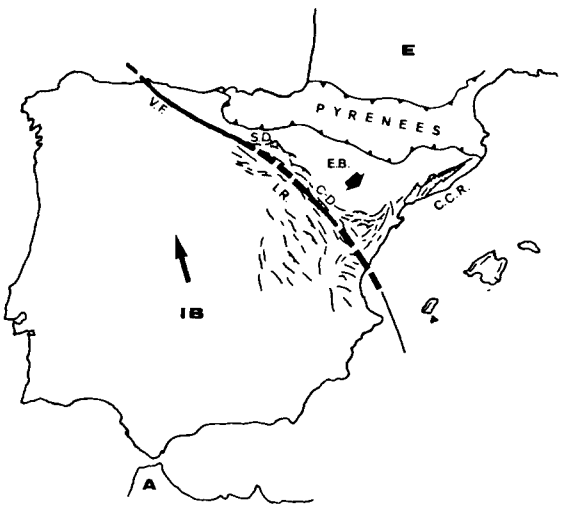


Fig. 14. Regional tectonic model (Model 1) to explain the clockwise rotation of the studied area. The NE part of Iberian plate is expelled towards the SE due to the northwards displacement of Iberia. Symbols: IB, Iberia; E, European plate; V.F., Ventaniella Fault; S.D., Sierra de la Demanda; C.D., Calatayud–Daroca Basin; E.B., Ebro Basin; I.R., Iberian Range; C.C.R., Catalan Coastal Range.

In model 1, the displacement towards the SE of the block would take place along a curved fault, leading to its clockwise rotation relative to the western part of Iberia during the Pyrenean compression. The block displacement would also have to be accommodated in the north somewhere in the present-day Pyrenees. Identifying this features within the known tectonic framework of the area is one of the key points when discussing model 1.

NW–SE faults along the Iberian Range were developed during a wrench faulting regime which affected southern Europe during the Late Hercynian (Arthaud and Matte, 1975). This NW–SE trend has been the principal axis of the Mesozoic basins in the Iberian Range since the early Triassic (Garrido and Villena, 1977; Canérot, 1985). The thickness and distribution of lower Triassic deposits and the localization of the basaltic emissions in the upper Triassic are some of the arguments that support the formation of these fault-controlled basins during Triassic–Liassic rifting (Alvaro et al., 1979).

During the compressional phase, from middle –upper Eocene to upper Oligocene, the major basement faults of the Iberian Range (NW–SE) moved reverse–dextral as consequence of a N–S regional compression. The result of the basement faults slip was folding and thrusting in the Mesozoic cover (Guimerà, 1984).

In that system of Iberian faults, the hypothetic fault, along which the 'Ebro block' would have slipped relative to the 'Stable Meseta', would correspond from NW to SE to the following features: the Ventaniella Fault (Julivert, 1965), the thrusts of the Sierra de la Demanda and the fault system of the Calatayud–Daroca Basin (Colomer, 1987).

Dextral movement along the above-mentioned faults is known to have occurred during the Paleogene. Tertiary rocks of the Ebro Basin are overthrust by the Sierra de la Demanda (Fig. 14). This E–W trending thrust is consistent with a dextral movement of the NW–SE faults. The

NW–SE faults at the SE margin of the Calatayud–Daroca Basin are well developed and marked by fault-gauge zones, which had an important dextral component (Colomer, 1987) during the Paleogene.

The continuation to the south of the suggested fault is difficult to recognize. There is no evidence in the area beyond the linking zone between the Iberian Range and the Catalan Coastal Range (Fig. 14). Nevertheless, the NNW–SSE trending fault between Alicante and Ibiza deduced from offshore geophysical studies (Mauffret, 1976), could correspond to the suggested structure.

The northern limit of the 'Ebro Block' remains as one of the geological problems of the model. Usually the North Pyrenean Fault is taken as the plate boundary and as a major feature in the Pyrenean evolution. However, according to model 1, the block limit may be hidden under the south Pyrenean thrust sheets. In fact, it may be argued that the North Pyrenean Fault was a major feature during the Cretaceous but not during the Paleogene.

An 'Ebro Block' has been described by Olivet (1978) and Malod (1982); however, the boundaries and the geological significance are different when compared with the proposed block in model 1.

7.2. Model 2

Obviously the geographical distribution of the samples is restricted and therefore the extrapolation of observations to the whole of the Ebro Basin is hypothetical. In addition, the large displacement implied by the curvature of the southern fault in model 1 is a major difficulty. Thus, an alternative model based on local structures could be considered.

As described before, the major structures of the Catalan Coastal Range are a system of right-stepping, en echelon, left-lateral strike-slip faults (Fig. 15A). Convergent sinistral slip on these faults took place between early Eocene and late Oligocene. In this strike-slip fault system, the SE fault blocks experienced an uplift relative to the NW ones. Associated with the sinistral slip, the fault-bounded blocks could have experienced a clockwise rotation due to the strike-slip displacement in the way

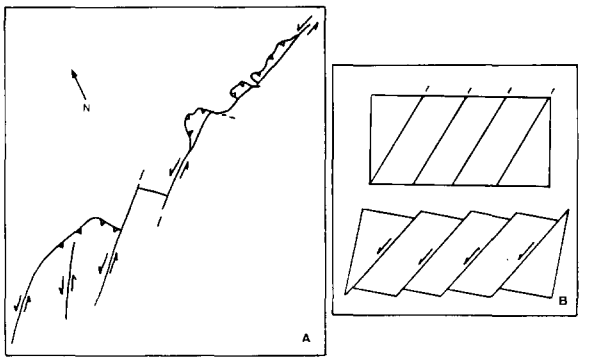


Fig. 15. Local tectonic model (Model 2). (A) Structural pattern of strike-slip faults along the SE margin of the Ebro Basin (Anadón et al., 1985). (B) Clockwise rotation of a set of faulted blocks during sinistral movement of the faults (based on Freund (1974) and Garfunkel (1974)). Note the analogies between this mechanism and the structural pattern of (A).

proposed by Freund (1974) and Garfunkel (1974) (Fig. 15B). In a simplified geometrical model of rigid fault blocks, these authors showed that in order to remain in contact with the boundary of the fault domain, the blocks must rotate and that the sense of rotation would be opposite to the sense of the fault slip. Moreover, the model of rigid blocks rotation based on the work of these authors, implies that the left-lateral, en echelon, strike-slip faults would be enclosed into a dextral 'megashear'.

From the discussion of models 1 and 2 it is clear that the establishment of either one of the models can be advanced only if further work is carried out on a broader scale in the Ebro Basin as well as in the Meseta and the Pyrenees. However, in the light of our knowledge today, we can conclude that the paleomagnetic data show that, at least in a sector NE of the so-called 'Stable Meseta', a clockwise rotation relative to the other areas of the Meseta has taken place. This implies that the Iberian Plate has not acted as a rigid block during the Paleogene.

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